

Optimization for Machine Learning *Outlook*

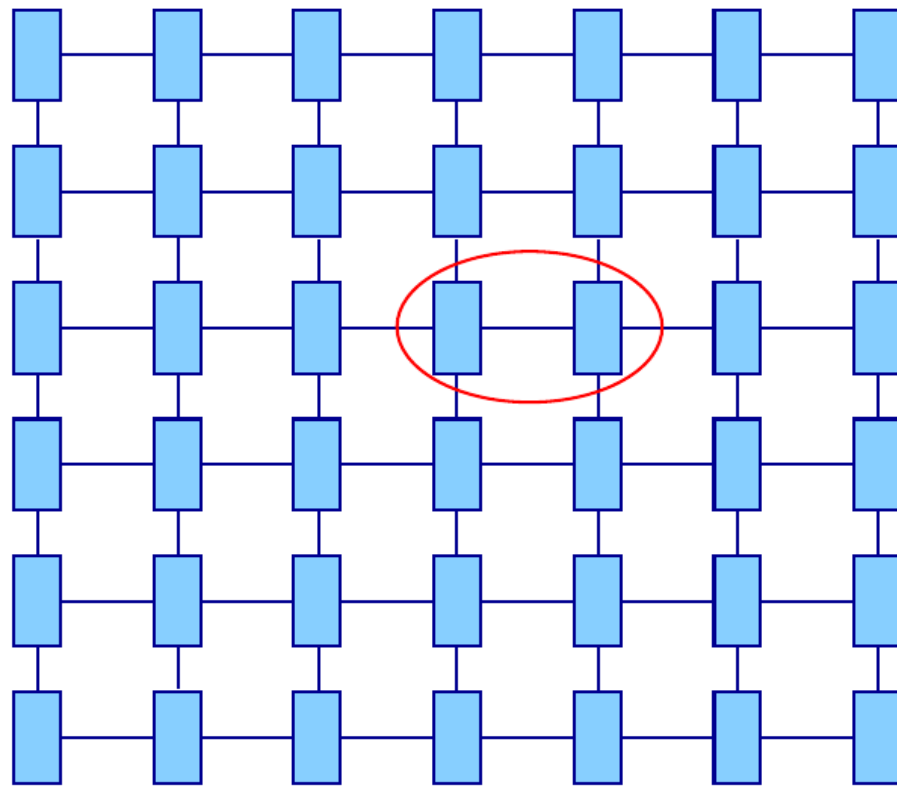
Bogdan Savchynskyy

What we learned

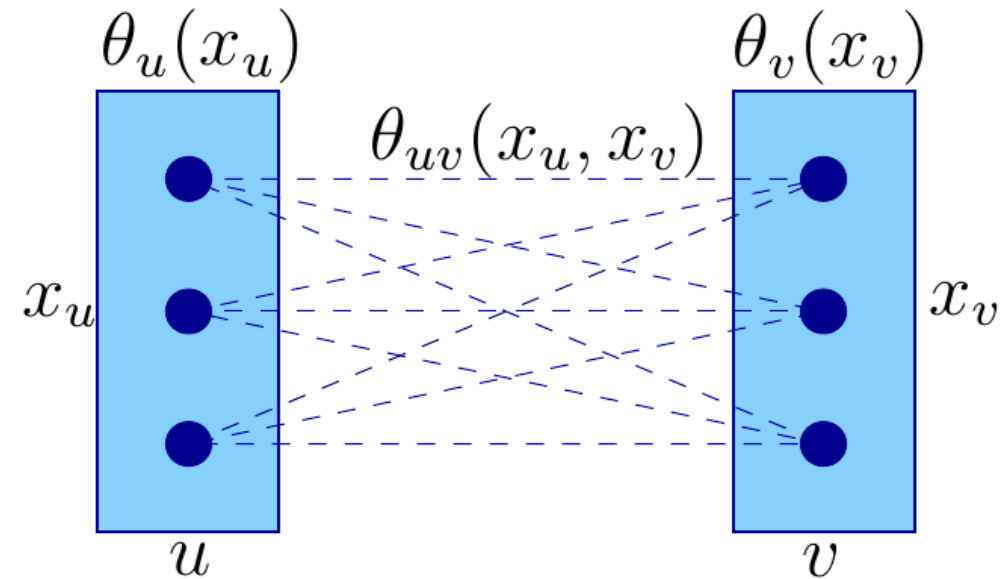
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Energy Minimization for Graphical Models

$$\bar{x}^* = \min_{\bar{x} \in \mathcal{X}_{\mathcal{V}}} E(\bar{x}) := \min_{\bar{x} \in \mathcal{X}_{\mathcal{V}}} \sum_{v \in \mathcal{V}} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v)$$



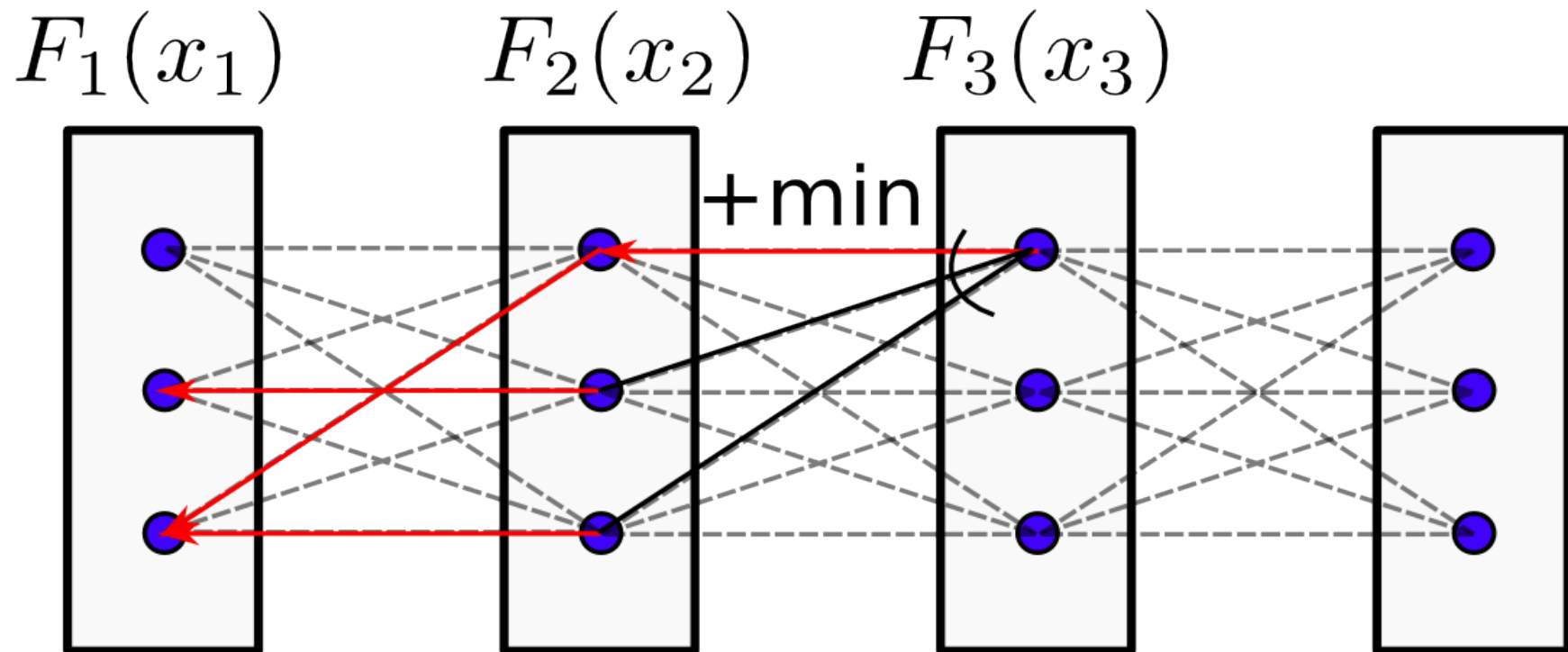
graph $(\mathcal{V}, \mathcal{E})$



Combinatorial optimization problem. Total enumeration has exponential complexity!

Dynamic Programming

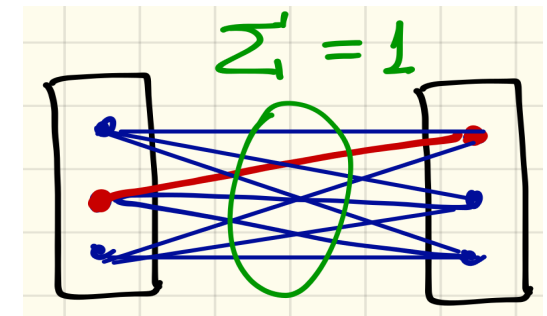
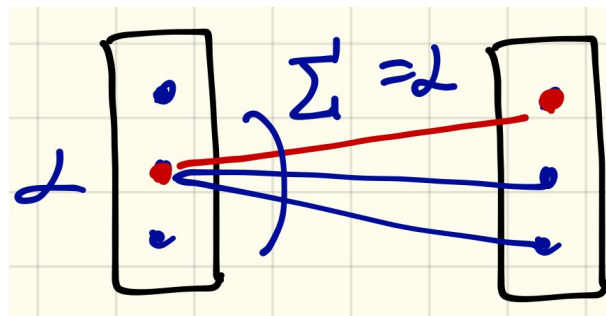
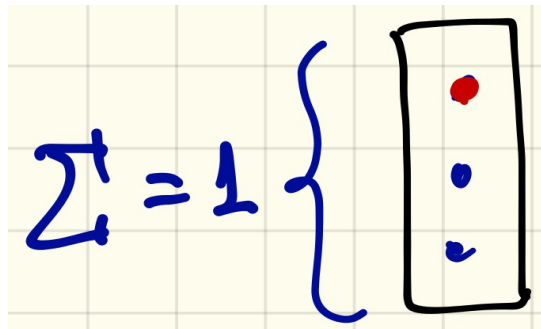
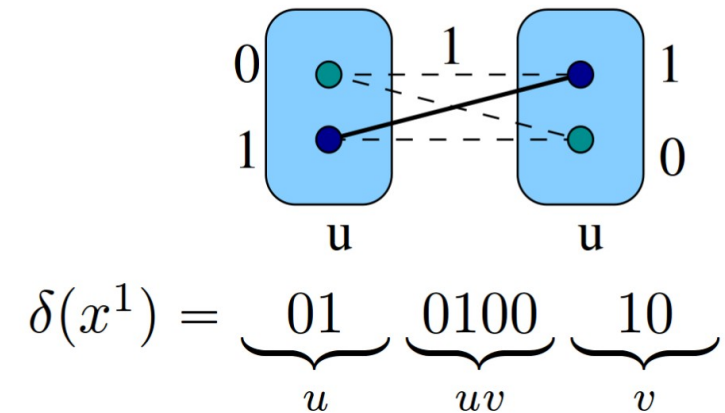
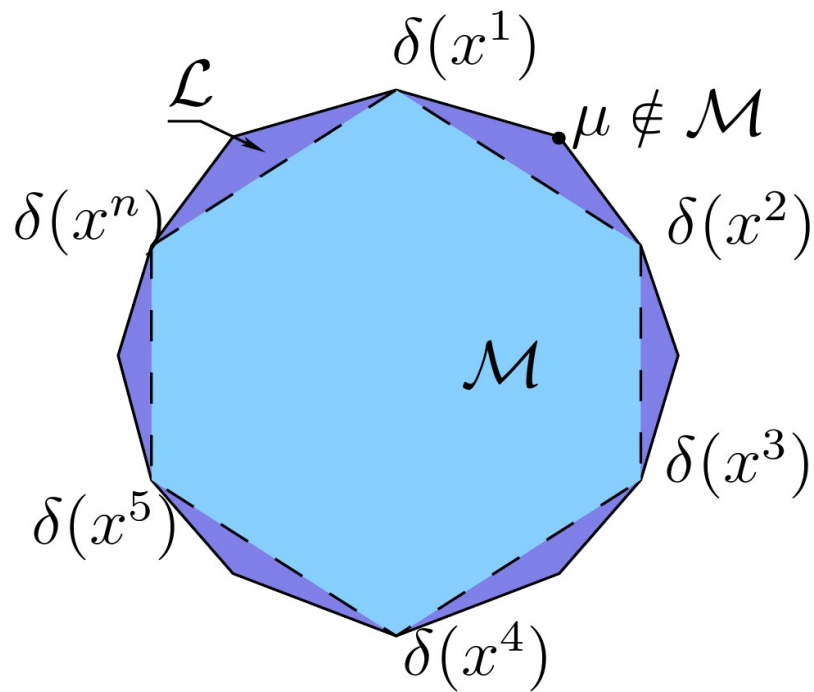
Compute the shortest path from the left:



$$F_i(s) := \min_{t \in i-1} F_{i-1}(t) + q_i(s, t)$$

Linear complexity of an algorithm for exact solution: for acyclic graphs!

(I)LP Formulations: Local and Marginal Polytopes



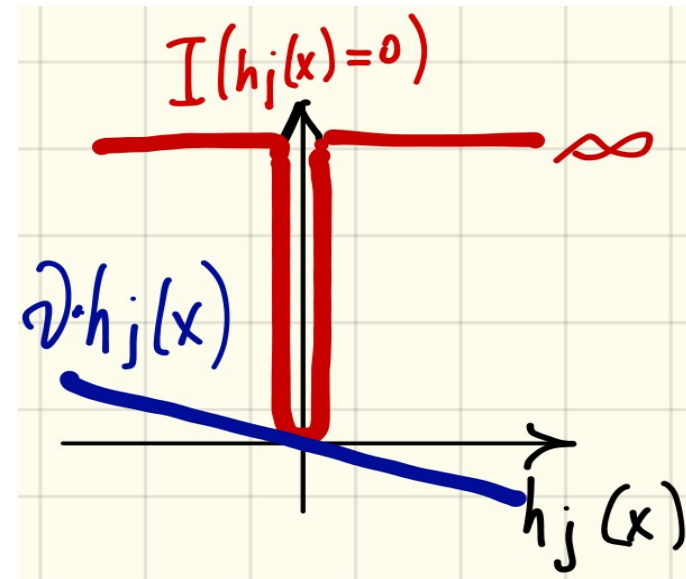
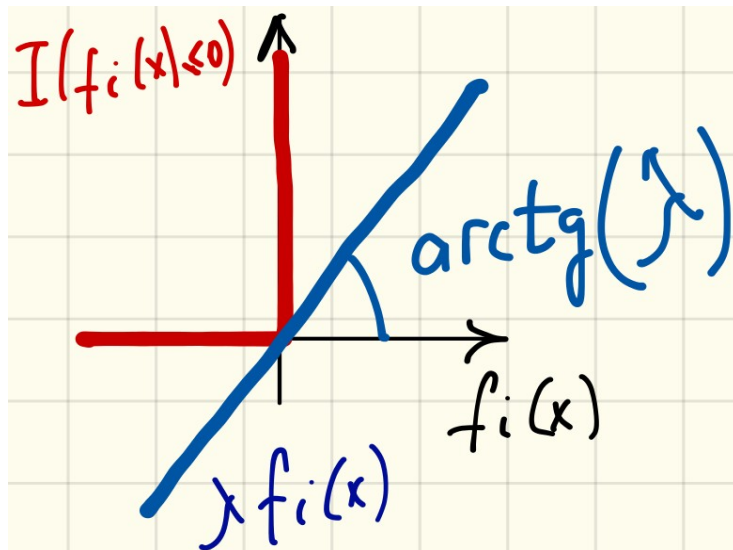
ILP formulation: out-of-the-shelf-solvers!

LP relaxation – polynomially solvable!

Lagrangian Dual: Concave Lower Bound

$$\tilde{f}(x) = f_0(x) + \sum_{i=1}^n I(f_i(x) \leq 0) + \sum_{j=1}^m I(h_j(x) = 0)$$

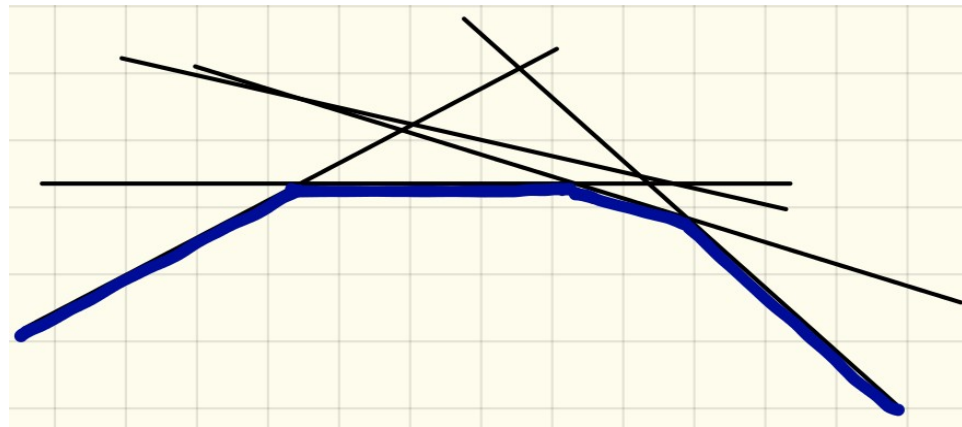
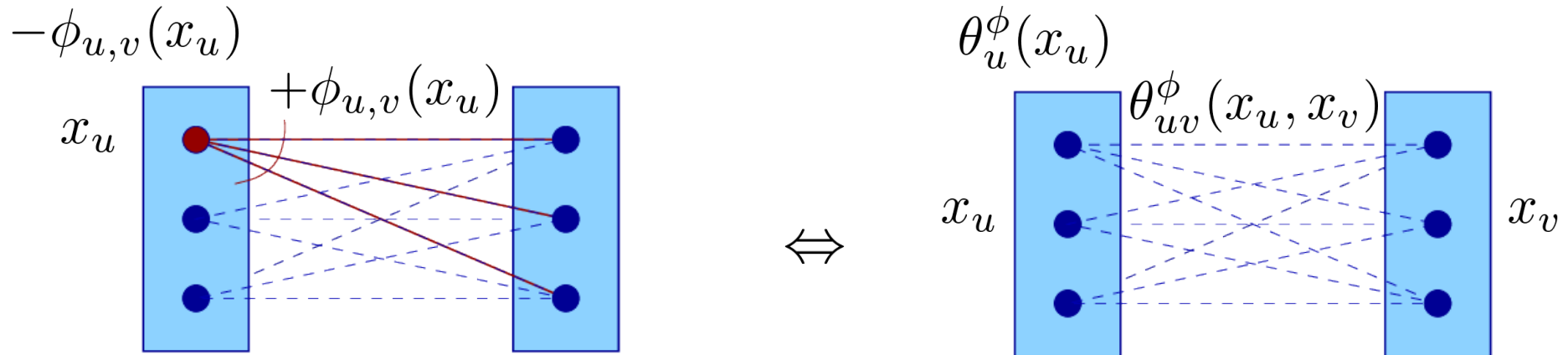
$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^n \lambda_i f_i(x) + \sum_{j=1}^m \nu_j h_j(x)$$



Construction of solvable (concave) lower bounds (relaxation) for constraint minimization!

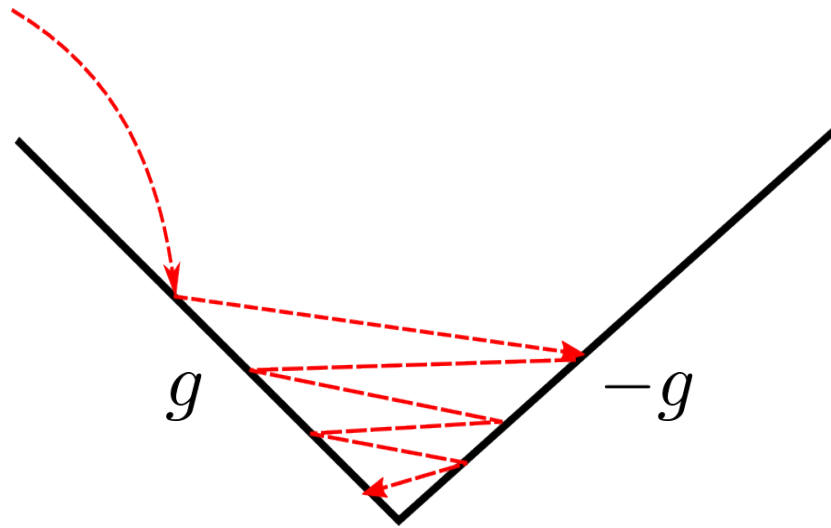
Dual LP: Reparametrization

Dual:
$$\max_{\phi} \left[D(\phi) := \sum_{v \in \mathcal{V}} \min_{s \in \mathcal{X}_v} \theta_v^{\phi}(x_v) + \sum_{uv \in \mathcal{E}} \min_{(s,t) \in \mathcal{X}_{uv}} \theta_{uv}^{\phi}(x_u, x_v) \right]$$

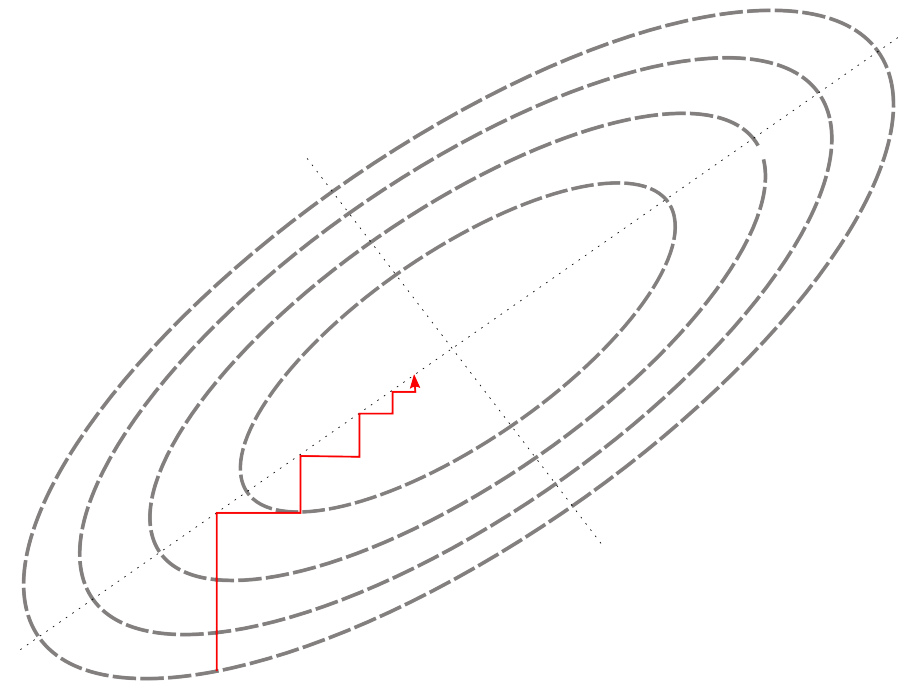


Concave unconstrained optimization ! Non-smooth :(

Basics of Convex Optimization



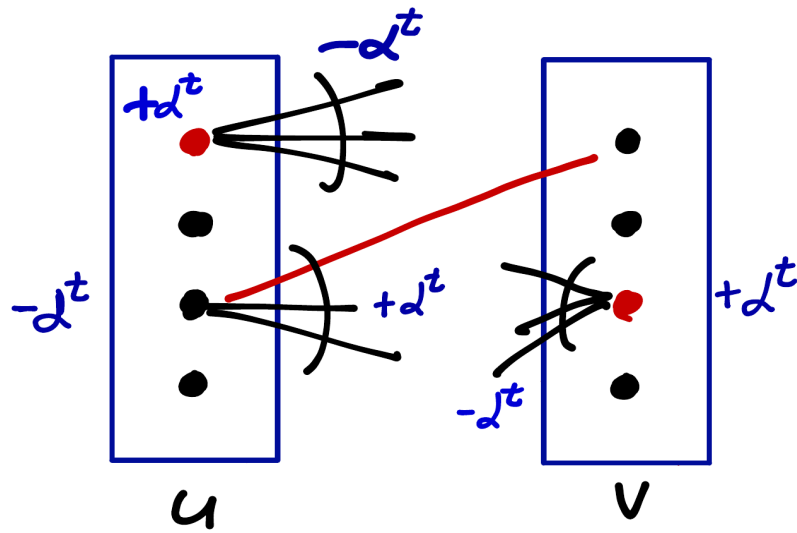
Subgradient method



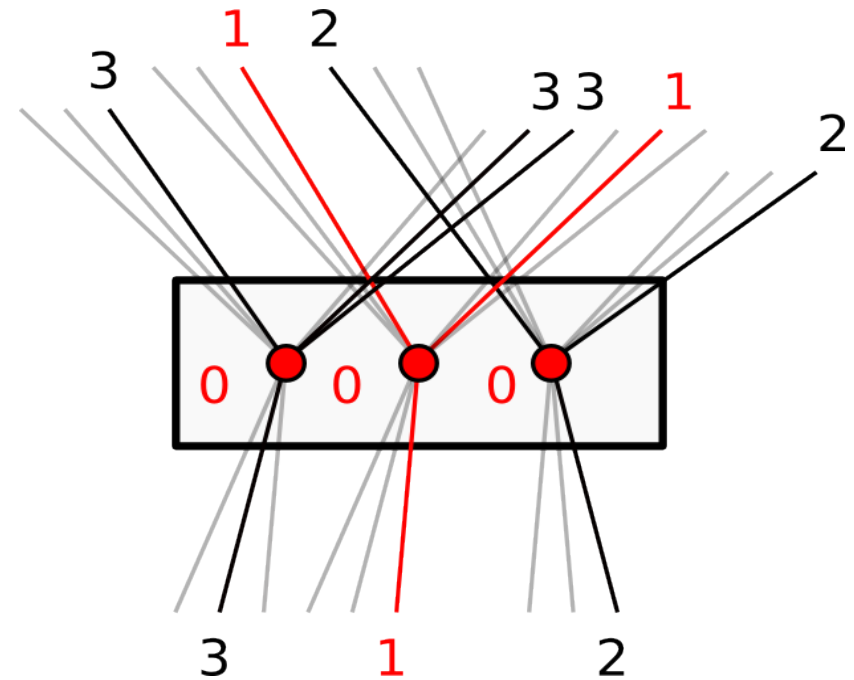
Coordinate descent

Simple scalable optimization methods.

Convex Optimization for Energy Minimization



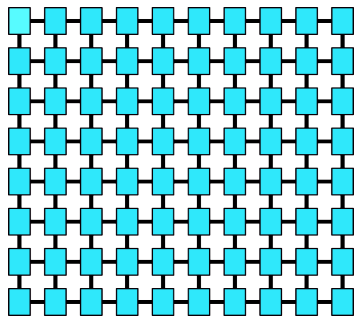
Subgradient method



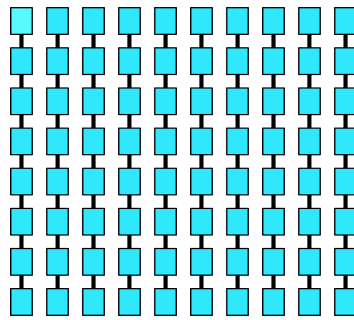
Diffusion Algorithm

Simple scalable optimization methods: Theoretical vs. Practical convergence

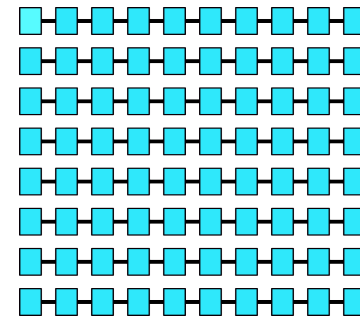
Lagrange Decomposition



=



+



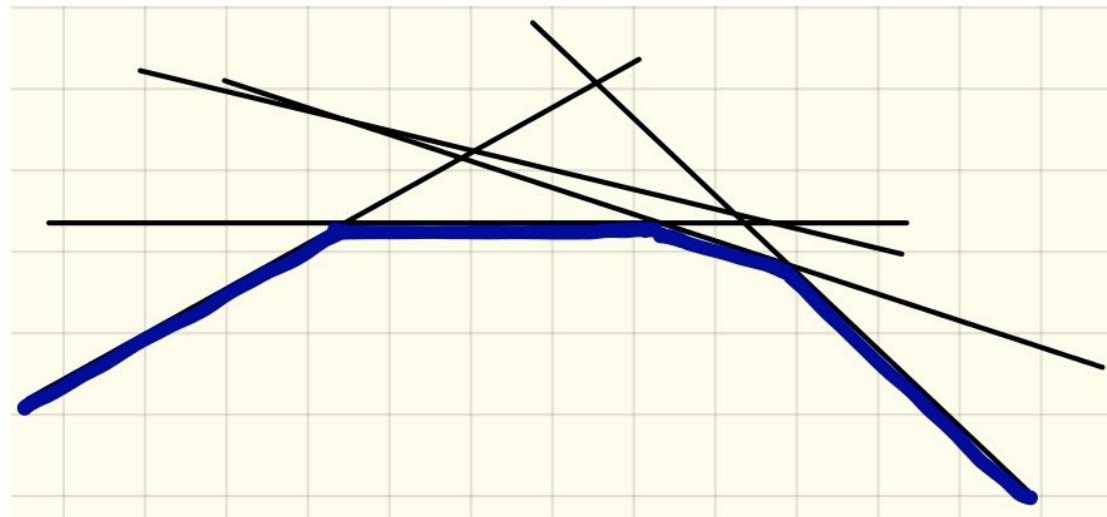
$$\langle \theta, \delta(x) \rangle$$

=

$$\langle \theta^c, \delta(x) \rangle$$

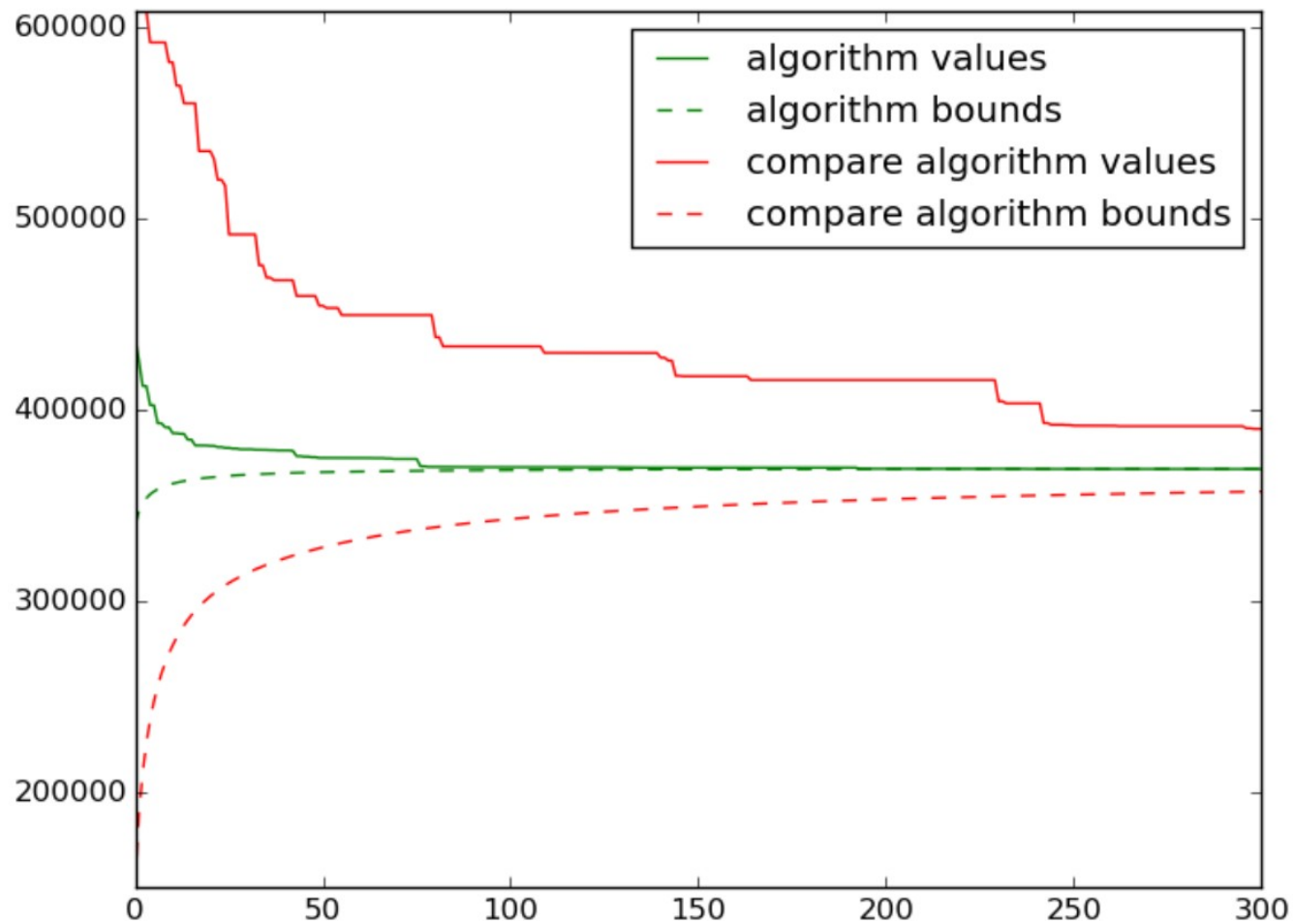
+

$$\langle \theta^r, \delta(x) \rangle$$



Another method to construct convex lower bounds.

Lagrange Decomposition



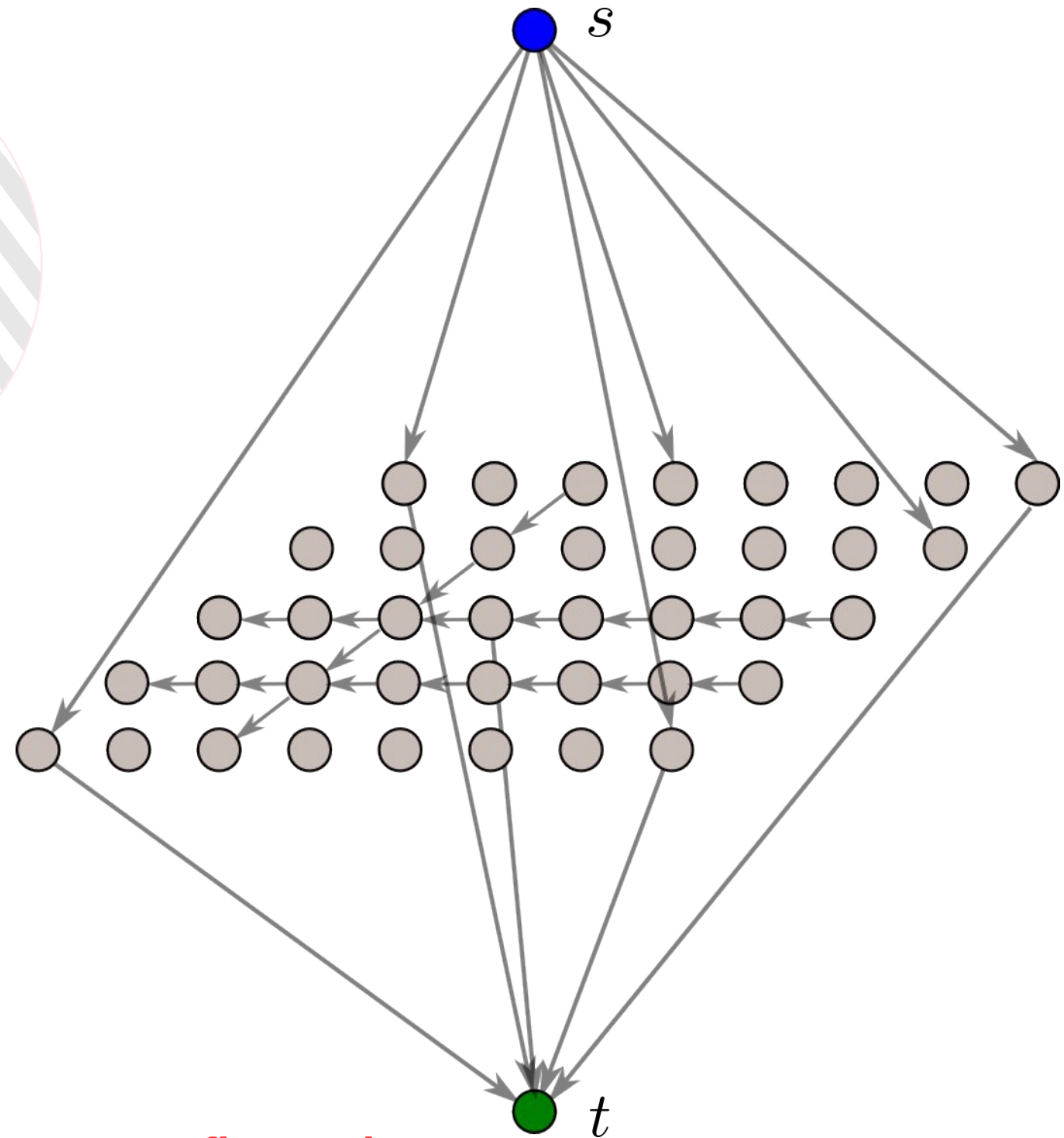
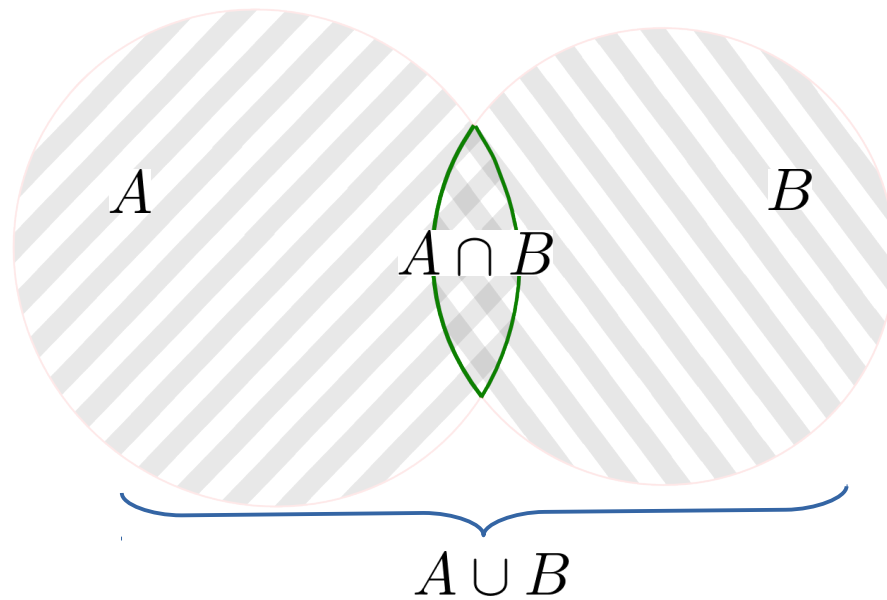
Row/column decomposition

Edge decomposition

Faster convergence!

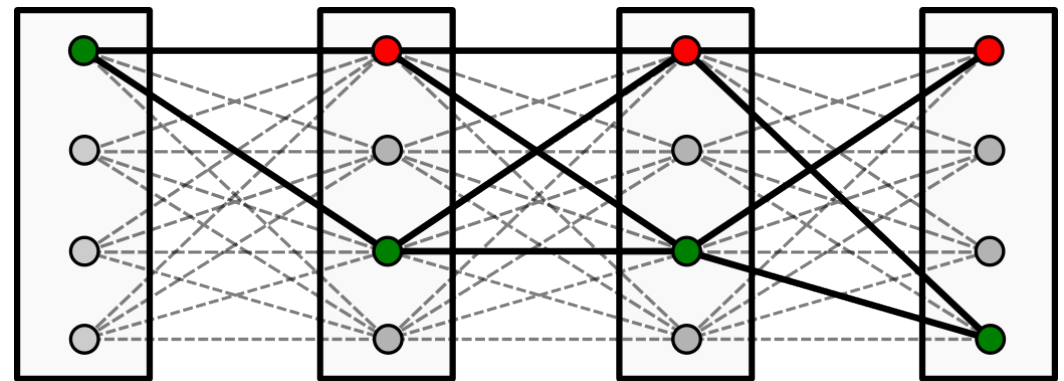
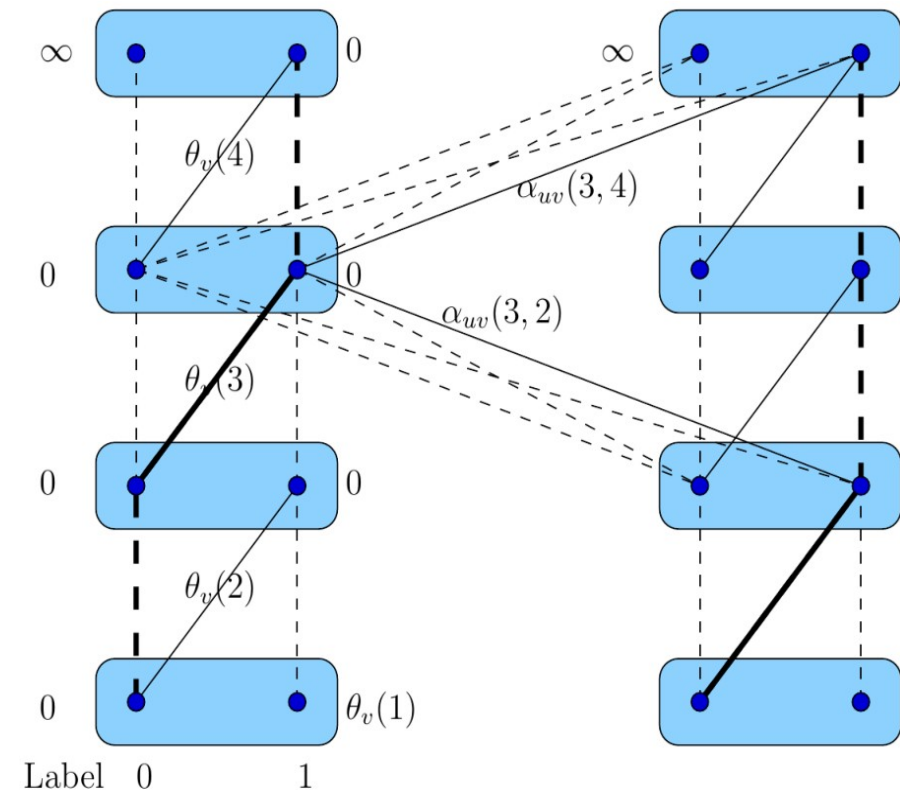
Binary Submodular Energy Minimization as Min-Cut

$$E(A) + E(B) \geq E(A \cap B) + E(A \cup B)$$



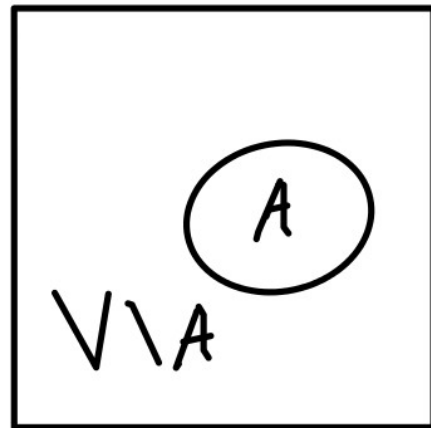
Fast and exact max-flow solver!

Multilabel to Binary Energy

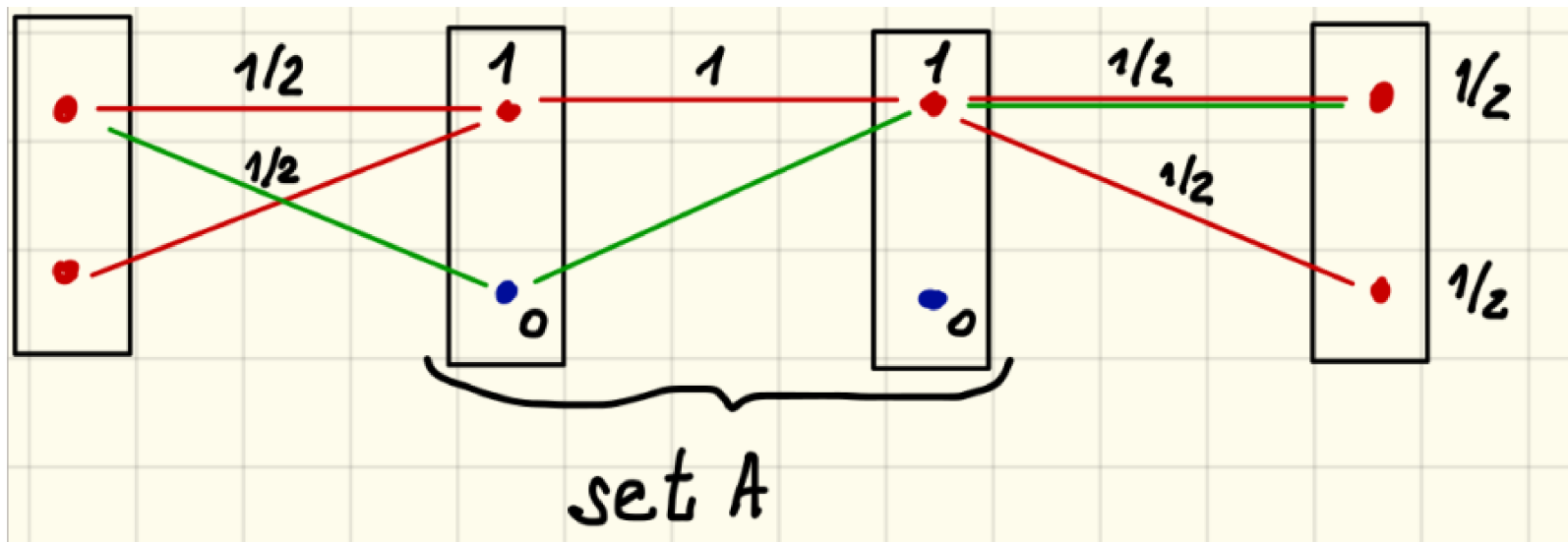


Move making algorithms, multilabel submodular energies solvable exactly!

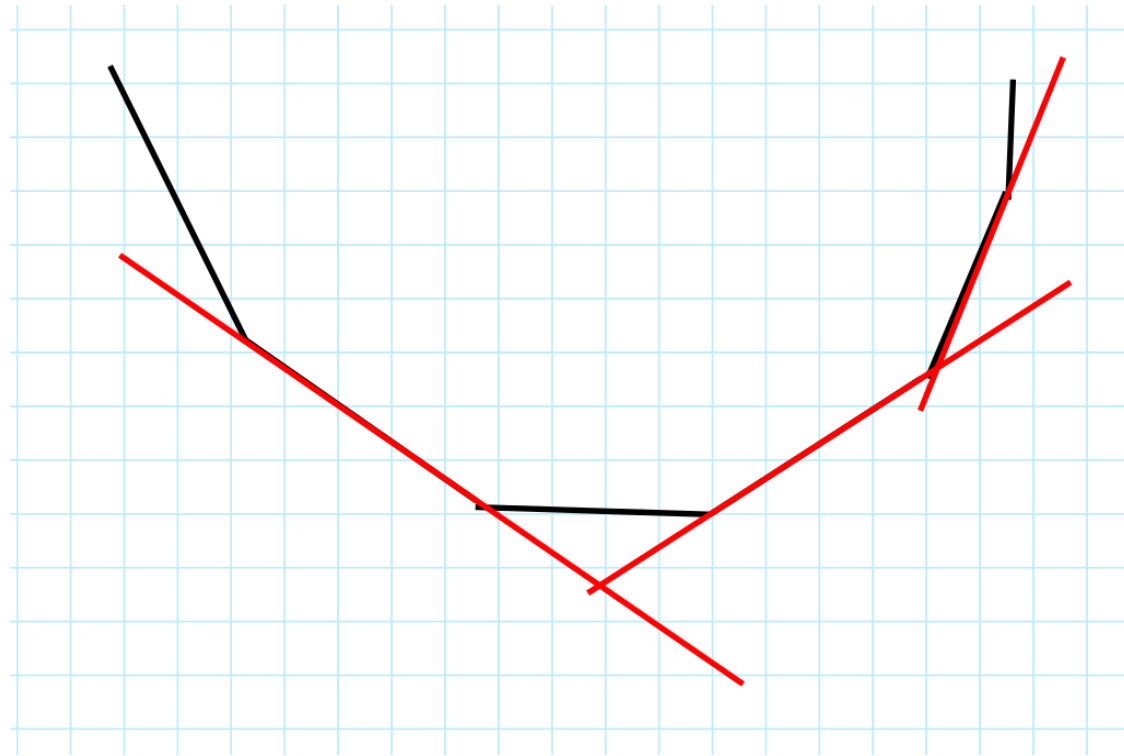
Binary LP Relaxation



A -integer Labels
 V/A -fractional Labels



Persistency, fusion moves – min-cut for arbitrary problems!



Cutting plane method

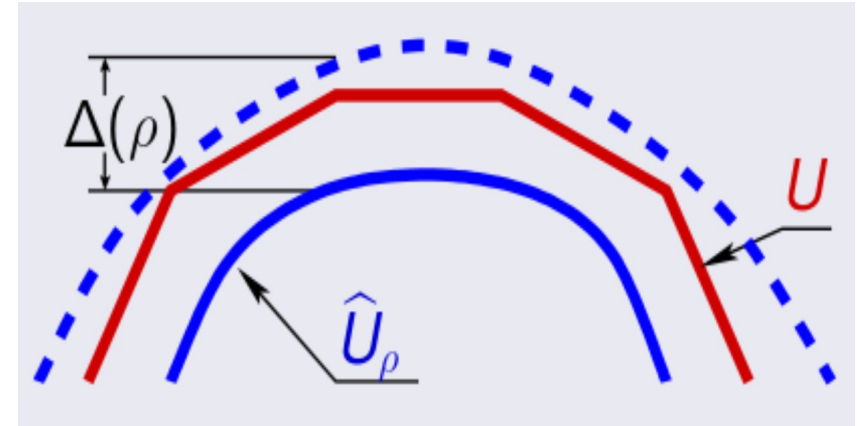
Parameter learning for graphical models !

Not Covered....

There are more thing to learn....

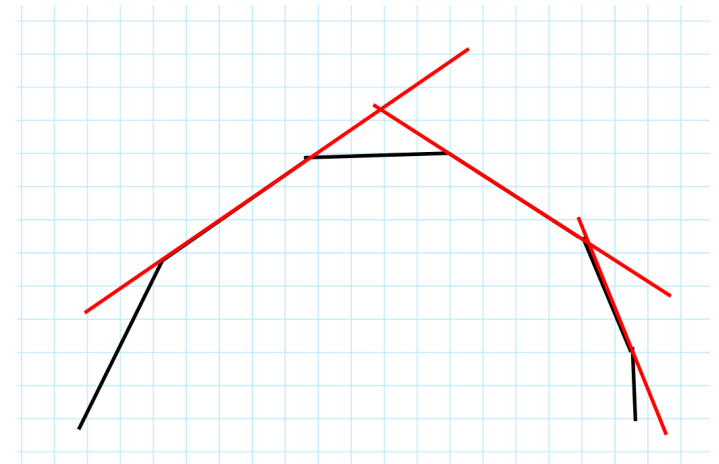
Non-smooth and combinatorial optimization

Smoothing methods

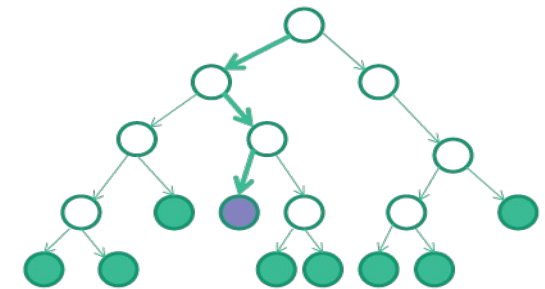


Bundle methods

$$\|w - w^t\| \rightarrow \min$$

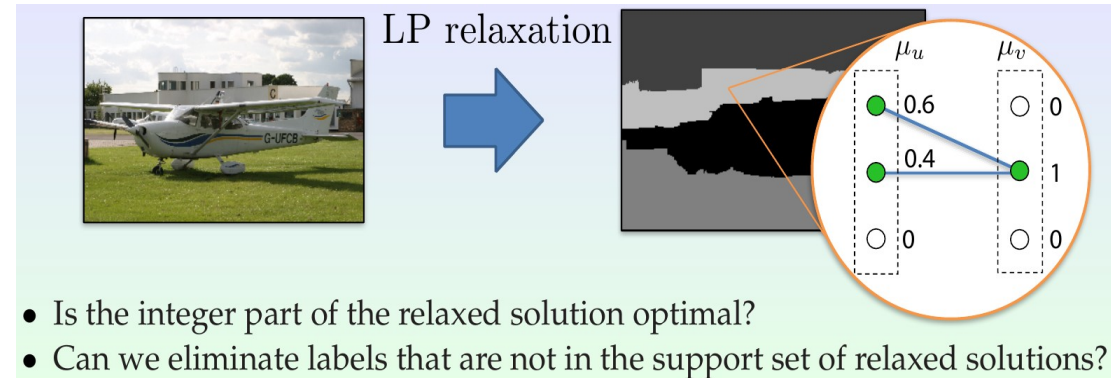


Branch and bound, combinatorial search

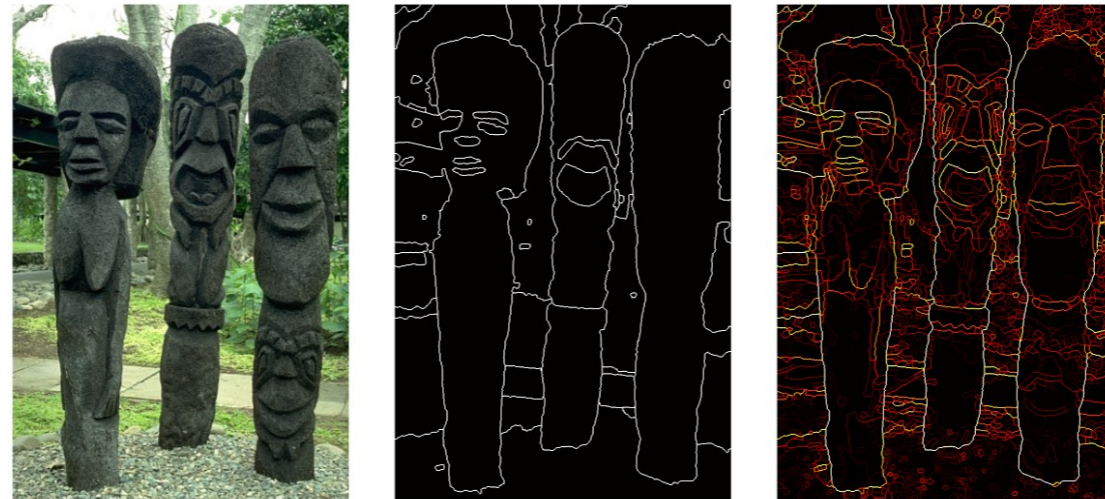


Each node in branch-and-bound is a new MIP

Partial optimality and meta-methods



Uncertainties in Solutions



Higher order models

...

Examination

Possible examination dates/time: August 8 – September 23, Mo, Wed.

To register email Katrin.Heber@tu-dre... and CC to me

Questions, feedback etc.: the same time, please **email me**
beforehand, if you want to come

To prepare to the exam use: script, slides, referenced literature.

Good by!

Thank you for your attention!